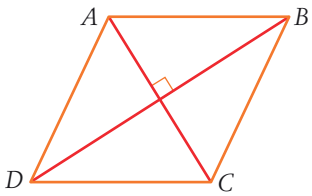
WORKSHEET

Chapter 1: Geometric proofs using vectors

1 Consider the parallelogram *ABCD* shown. Given that *AC* and *DB* are perpendicular, complete the missing parts of the following proof that *ABCD* is a rhombus.

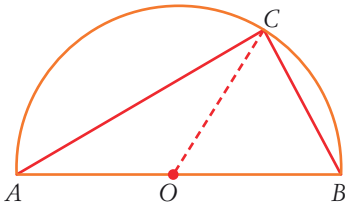


Proof

Opposite sides are equal and parallel.	$\mathbf{AB} = \ldots$ and $\mathbf{AD} = \ldots$
AC and *DB* are perpendicular.	$\mathbf{AC} \cdot \mathbf{DB} = \ldots$
Express **AC** and **DB** in terms of the sides of *ABCD*.	$(\mathbf{AB} + \ldots) \cdot (\mathbf{BC} + \ldots) = \ldots$
Use the fact that $\mathbf{AB} = \mathbf{DC}$, so $\mathbf{AB} = -\mathbf{CD}$.	$(\ldots + \ldots) \cdot (\mathbf{BC} + \ldots) = \ldots$
Expand the dot product.	$\ldots \cdot \mathbf{BC} - \ldots \cdot \mathbf{CD} + \mathbf{BC} \cdot \ldots + \mathbf{BC} \cdot \ldots = \ldots$
Simplify.	$-\lvert\ldots\rvert^2 + \lvert\mathbf{BC}\rvert^2 = \ldots$
Rearrange.	$\lvert\mathbf{BC}\rvert^2 = \lvert\ldots\rvert^2$
Take square roots.	$\lvert\mathbf{BC}\rvert = \lvert\ldots\rvert$
State the conclusion.	… sides are equal, so *ABCD* is a …
	QED

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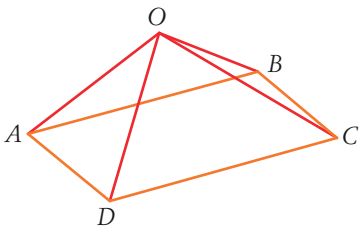
2 The diagram below shows a semicircle with centre O and a diameter with end points A and B. C is any point on the circumference. Use the dot product of **CA** and **CB** to prove that an angle inscribed in a semicircle is a right angle.



3 Two of the opposite sides of a quadrilateral are parallel and equal in length. You are required to prove that the quadrilateral is a parallelogram. Complete the missing parts of the proof shown below using position vectors.

Proof

Draw a diagram of quadrilateral $ABCD$.



Let O be the origin.

Let $AB = CD$ and $AB \parallel CD$.

AB and CD are equal and parallel. $\mathbf{AB} = \ldots$ [1]

Write **AB** as a sum of vectors. $\mathbf{OA} + \mathbf{AB} = \ldots$

Rearrange. $\mathbf{AB} = \ldots - \mathbf{OA}$

Write **DC** as a sum of vectors. $\mathbf{OD} + \mathbf{DC} = \ldots$

Rearrange. $\mathbf{DC} = \ldots - \ldots$

Substitute for **AB** and **DC** in [1]. $\ldots - \mathbf{OA} = \ldots - \ldots$

Add **OD** and subtract **OB** from both sides. $\ldots - \mathbf{OA} = \ldots - \ldots$ [2]

Use the fact that $\mathbf{OA} + \mathbf{AD} = \mathbf{OD}$. $\mathbf{AD} = \ldots - \ldots$

Use the fact that $\mathbf{OB} + \mathbf{BC} = \mathbf{OC}$. $\mathbf{BC} = \ldots - \ldots$

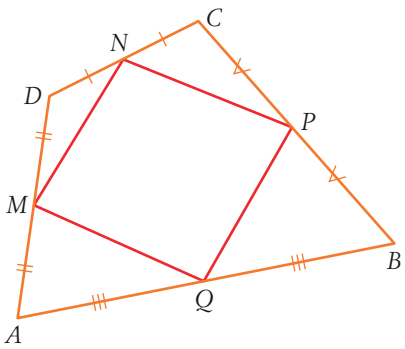
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Substitute for **AD** and **BC** in [2].	… = …
State the conclusion.	AD = … and AD || …
Opposite sides of $ABCD$ are equal and parallel.	… is a parallelogram.
	QED

4 The midpoints of the sides of a quadrilateral are joined. You are required to prove that the quadrilateral formed by joining these midpoints is a parallelogram. Complete the missing parts of the proof shown below using position vectors.

Proof

Draw a diagram.



Mark in M, N, P and Q such that:

$\mathbf{AM} = \mathbf{MD} = \frac{1}{2}\mathbf{AD}$

$\mathbf{DN} = \mathbf{NC} = \frac{1}{2}\mathbf{DC}$

$\mathbf{CP} = \mathbf{PB} = \frac{1}{2}\mathbf{CB}$

$\mathbf{AQ} = \mathbf{QB} = \frac{1}{2}\mathbf{AB}$

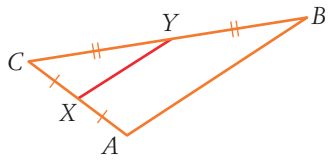
Let $\mathbf{AD} = \mathbf{a}$, $\mathbf{DC} = \mathbf{b}$, $\mathbf{CB} = \mathbf{c}$ and $\mathbf{BA} = \mathbf{d}$

Use head to tail addition of vectors.	$\mathbf{a} + \mathbf{b} + \ldots + \ldots = 0$
Rearrange.	$\mathbf{a} + \mathbf{b} + \ldots = -\ldots$
Write **MN** as the sum of vectors.	$\mathbf{MN} = \mathbf{MD} + \ldots$
Use $\mathbf{MD} = \frac{1}{2}\mathbf{AD}$ and $\mathbf{DN} = \frac{1}{2}\mathbf{DC}$.	$= \frac{1}{2}\ldots + \frac{1}{2}\ldots$

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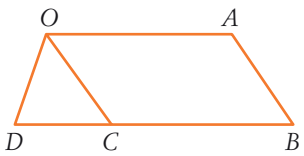
Use $\mathbf{AD} = \mathbf{a}$ and $\mathbf{DC} = \mathbf{b}$.	$= \frac{1}{2}\mathbf{a} + \frac{1}{2}\ldots$
Simplify.	$= \frac{1}{2}(\ldots + \ldots)$
Write $\mathbf{QP}$ as the sum of vectors.	$\mathbf{QP} = \mathbf{QB} + \ldots$
Use $\mathbf{QB} = \frac{1}{2}\mathbf{AB}$ and $\mathbf{PB} = \frac{1}{2}\mathbf{CB}$.	$= \frac{1}{2}\mathbf{AB} - \frac{1}{2}\ldots$
Use $\mathbf{BA} = \mathbf{d}$ and $\mathbf{CB} = \mathbf{c}$.	$= -\frac{1}{2}\mathbf{d} - \frac{1}{2}\ldots$
Simplify.	$= \frac{1}{2}(-\mathbf{d} - \ldots)$
Substitute for $-\mathbf{d}$.	$= \frac{1}{2}(\ldots + \ldots + \ldots - \ldots)$
Simplify.	$= \frac{1}{2}(\ldots + \ldots)$
Write the conclusion.	$\mathbf{MN} = \ldots$ so $MN = \ldots$ and $MN \parallel \ldots$
Use a similar process.	$\mathbf{QM} = \ldots$ so $QM = \ldots$ and $QM \parallel \ldots$
	… is a parallelogram.
	QED

5 In ΔABC shown, X is the midpoint of CA and Y is the midpoint of CB.
Prove that $\mathbf{XY} = \frac{1}{2}\mathbf{AB}$



6 Prove that the sum of the squares of the lengths of the diagonals of a parallelogram is equal to the squares of the lengths of all sides.

7 $OABC$ is a parallelogram. BC is produced to D such that $DC : CB = 1 : 2$



Prove that $\mathbf{DB} = \frac{5}{2}\mathbf{a} - \mathbf{b}$.

8 Consider the semicircle with centre (0, 0) and radius 1. Prove that the lines connecting any point on the semicircle with the points (1,0) and $(-1, 0)$ are perpendicular.

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Answers

1 $\mathbf{AB} = \mathbf{DC}$ and $\mathbf{AD} = \mathbf{BC}$

$$\mathbf{AC} \cdot \mathbf{DB} = 0$$
$$(\mathbf{AB} + \mathbf{BC}) \cdot (\mathbf{BC} + \mathbf{CD}) = 0$$
$$(-\mathbf{CD} + \mathbf{BC}) \cdot (\mathbf{BC} + \mathbf{CD}) = 0$$
$$-\mathbf{CD} \cdot \mathbf{BC} - \mathbf{CD} \cdot \mathbf{CD} + \mathbf{BC} \cdot \mathbf{BC} + \mathbf{BC} \cdot \mathbf{CD} = 0$$
$$-|\mathbf{CD}|^2 + |\mathbf{BC}|^2 = 0$$
$$|\mathbf{BC}|^2 = |\mathbf{CD}|$$
$$|\mathbf{BC}| = |\mathbf{CD}|$$

Adjacent sides are equal, so *ABCD* is a rhombus.

QED

2 Find the dot product of **CA** and **CB**.

$$\mathbf{CA} \cdot \mathbf{CB} = (\mathbf{OA} - \mathbf{OC}) \cdot (\mathbf{OB} - \mathbf{OC})$$

OA and **OB** are radii, so $|\mathbf{OA}| = |\mathbf{OB}|$.

$$\mathbf{OB} = -\mathbf{OA}$$

Substitute for **OB**.

$$\mathbf{CA} \cdot \mathbf{CB} = (\mathbf{OA} - \mathbf{OC}) \cdot (-\mathbf{OA} - \mathbf{OC})$$

Expand the dot product.

$$= -\mathbf{OA} \cdot \mathbf{OA} - \mathbf{OA} \cdot \mathbf{OC} + \mathbf{OC} \cdot \mathbf{OA} + \mathbf{OC} \cdot \mathbf{OC}$$

Simplify.

$$= -|\mathbf{OA}|^2 + |\mathbf{OC}|^2$$

OA and **OC** are radii.

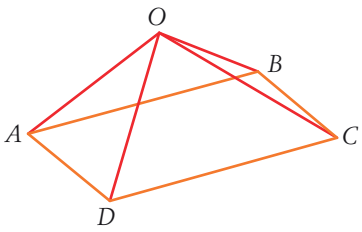
$$= 0$$

So **CA** and **CB** are perpendicular.

QED

3 Proof

Draw a diagram of quadrilateral *ABCD*.



Let *O* be the origin.

Let $AB = CD$ and $AB \parallel CD$.

AB and *CD* are equal and parallel.

$$\mathbf{AB} = \mathbf{DC} \quad [1]$$

Write **AB** as a sum of vectors.

$$\mathbf{OA} + \mathbf{AB} = \mathbf{OB}$$

Rearrange.

$$\mathbf{AB} = \mathbf{OB} - \mathbf{OA}$$

Write **DC** as a sum of vectors.

$$\mathbf{OD} + \mathbf{DC} = \mathbf{OC}$$

Rearrange.

$$\mathbf{DC} = \mathbf{OC} - \mathbf{OD}$$

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Substitute for **AB** and **DC** in [1]. $\mathbf{OB} - \mathbf{OA} = \mathbf{OC} - \mathbf{OD}$

Add **OD** and subtract **OB** from both sides. $\mathbf{OD} - \mathbf{OA} = \mathbf{OC} - \mathbf{OB}$ [2]

Use the fact that $\mathbf{OA} + \mathbf{AD} = \mathbf{OD}$. $\mathbf{AD} = \mathbf{OD} - \mathbf{OA}$

Use the fact that $\mathbf{OB} + \mathbf{BC} = \mathbf{OC}$. $\mathbf{BC} = \mathbf{OC} - \mathbf{OB}$

Substitute for **AD** and **BC** in [2]. $\mathbf{AD} = \mathbf{BC}$

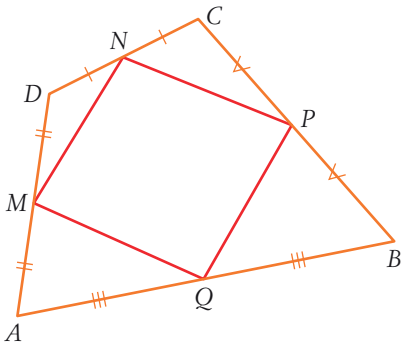
State the conclusion. $AD = BC$ and $AD \parallel BC$

Opposite sides of *ABCD* are equal and parallel. *ABCD* is a parallelogram.

QED

4 Proof

Draw a diagram.



Mark in *M*, *N*, *P* and *Q* such that:

$\mathbf{AM} = \mathbf{MD} = \frac{1}{2}\mathbf{AD}$

$\mathbf{DN} = \mathbf{NC} = \frac{1}{2}\mathbf{DC}$

$\mathbf{CP} = \mathbf{PB} = \frac{1}{2}\mathbf{CB}$

$\mathbf{AQ} = \mathbf{QB} = \frac{1}{2}\mathbf{AB}$

Let $\mathbf{AD} = \mathbf{a}$, $\mathbf{DC} = \mathbf{b}$, $\mathbf{CB} = \mathbf{c}$ and $\mathbf{BA} = \mathbf{d}$

Use head to tail addition of vectors. $\mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d} = 0$

Rearrange. $\mathbf{a} + \mathbf{b} + \mathbf{c} = -\mathbf{d}$

Write **MN** as the sum of vectors. $\mathbf{MN} = \mathbf{MD} + \mathbf{DN}$

Use $\mathbf{MD} = \frac{1}{2}\mathbf{AD}$ and $\mathbf{DN} = \frac{1}{2}\mathbf{DC}$. $= \frac{1}{2}\mathbf{AD} + \frac{1}{2}\mathbf{DC}$

Use $\mathbf{AD} = \mathbf{a}$ and $\mathbf{DC} = \mathbf{b}$.	$= \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b}$
Simplify.	$= \frac{1}{2}(\mathbf{a} + \mathbf{b})$
Write $\mathbf{QP}$ as the sum of vectors.	$\mathbf{QP} = \mathbf{QB} + \mathbf{BP}$
Use $\mathbf{QB} = \frac{1}{2}\mathbf{AB}$ and $\mathbf{PB} = \frac{1}{2}\mathbf{CB}$.	$= \frac{1}{2}\mathbf{AB} - \frac{1}{2}\mathbf{CB}$
Use $\mathbf{BA} = \mathbf{d}$ and $\mathbf{CB} = \mathbf{c}$.	$= -\frac{1}{2}\mathbf{d} - \frac{1}{2}\mathbf{c}$
Simplify.	$= \frac{1}{2}(-\mathbf{d} - \mathbf{c})$
Substitute for $-\mathbf{d}$.	$= \frac{1}{2}(\mathbf{a} + \mathbf{b} + \mathbf{c} - \mathbf{c})$
Simplify.	$= \frac{1}{2}(\mathbf{a} + \mathbf{b})$
Write the conclusion.	$\mathbf{MN} = \mathbf{QP}$ so $MN = QP$ and $MN \parallel QP$
Use a similar process.	$\mathbf{QM} = \mathbf{PN}$ so $QM = PN$ and $QM \parallel PN$
	$MNPQ$ is a parallelogram.
	QED

5 Assume O is the origin and

$\mathbf{OA} = \mathbf{a}$, $\mathbf{OB} = \mathbf{b}$, $\mathbf{OC} = \mathbf{c}$, $\mathbf{OX} = \mathbf{x}$ and $\mathbf{OY} = \mathbf{y}$

Proof

$$\mathbf{x} = \frac{1}{2}(\mathbf{a} + \mathbf{c})$$

$$\mathbf{y} = \frac{1}{2}(\mathbf{c} + \mathbf{b})$$

$$\mathbf{x} - \mathbf{y} = \frac{1}{2}\mathbf{c} + \frac{1}{2}\mathbf{b} - \frac{1}{2}\mathbf{a} - \frac{1}{2}\mathbf{c}$$

$$= \frac{1}{2}\mathbf{b} - \frac{1}{2}\mathbf{a}$$

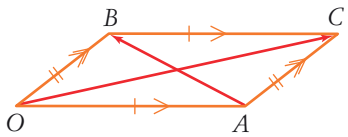
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$= \frac{1}{2}(\mathbf{b} - \mathbf{a})$

$\mathbf{YX} = \frac{1}{2}\mathbf{BA}$

QED

6 Draw a diagram.



Proof

Let $\mathbf{OB} = \mathbf{b}$ and $\mathbf{OA} = \mathbf{a}$.

$\mathbf{OC} = \mathbf{a} + \mathbf{b}$

$\mathbf{a} + \mathbf{AB} = \mathbf{b}$

$\mathbf{AB} = \mathbf{b} - \mathbf{a}$

Sum of squares of diagonals $= |\mathbf{OC}|^2 + |\mathbf{AB}|^2$

$= |\mathbf{a} + \mathbf{b}|^2 + |\mathbf{b} - \mathbf{a}|^2$

$= (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b}) + (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b})$ (Using the fact that $|\mathbf{a}|^2 = \mathbf{a} \cdot \mathbf{a}$.)

$= \mathbf{a} \cdot \mathbf{a} + 2\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{b} - 2\mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{a}$

$= 2\mathbf{a} \cdot \mathbf{a} + 2\mathbf{b} \cdot \mathbf{b}$

$= 2|\mathbf{a}|^2 + 2|\mathbf{b}|^2$

QED

7 Assume O is the origin and:

$\mathbf{OA} = \mathbf{a}$ and $\mathbf{OB} = \mathbf{b}$

Proof

$\mathbf{DA} = \mathbf{DB} + \mathbf{BA}$

$\mathbf{DB} = \mathbf{DC} + \mathbf{CB}$

$\mathbf{DC} = \frac{1}{2}\mathbf{a}$

$\mathbf{CB} = \mathbf{a}$

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So, $\mathbf{DB} = \frac{1}{2}\mathbf{a} + \mathbf{a} = \frac{3}{2}\mathbf{a}$

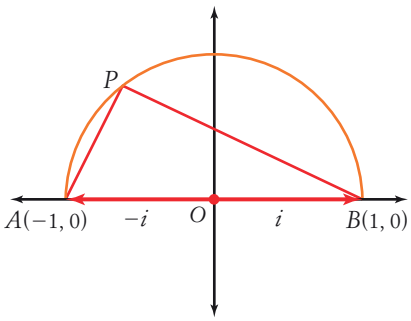
$\mathbf{BA} = \mathbf{a} - \mathbf{b}$

Substitute for $\mathbf{DB} + \mathbf{BA}$. $\mathbf{DA} = \frac{3}{2}\mathbf{a} + \mathbf{a} - \mathbf{b}$

$= \frac{5}{2}\mathbf{a} - \mathbf{b}$

QED

8 Draw a diagram.



Proof

$\mathbf{OP} = \mathbf{p}$, $\mathbf{OA} = -\mathbf{i}$ and $\mathbf{OB} = \mathbf{i}$

$\mathbf{PA} = \mathbf{p} - (-\mathbf{i})$

$= \mathbf{p} + \mathbf{i}$

$\mathbf{BP} = \mathbf{p} - \mathbf{i}$

$\mathbf{PA} \cdot \mathbf{BP} = (\mathbf{p} + \mathbf{i}) \cdot (\mathbf{p} - \mathbf{i})$

$= \mathbf{p} \cdot \mathbf{p} - \mathbf{p} \cdot \mathbf{i} + \mathbf{i} \cdot \mathbf{p} - \mathbf{i} \cdot \mathbf{i}$

$= \mathbf{p} \cdot \mathbf{p} - \mathbf{i} \cdot \mathbf{i}$

$= |\mathbf{p}|^2 - 1$

But $|\mathbf{p}| = 1$

So, $\mathbf{PA} \cdot \mathbf{BP} = 0$

So, **PA** and **BP** are perpendicular.

QED